# Critic-Free Group-Relative Policy Optimization for Anti-Sway Control of a Double-Pendulum Crane

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GitHub: https://github.com/Zephyr-sue/rl_crane_antisway

Abstract—We formulate anti-sway transport with an overhead crane, where a trolley carries a hook and a payload (a double pendulum), as a continuous-control RL problem. Payload mass, cable lengths, initial sway and wind gusts are randomized. We adapt Group Relative Policy Optimization (GRPO) [1], originally proposed for LLM reasoning, to physical control. GRPO has no value network: $G$ rollouts replay the same randomized scenario on a GPU-batched simulator and serve as each other's baseline. We evaluate on 8 test conditions, including out-of-distribution plants and doubled gusts. GRPO succeeds in 93–97% of random scenarios across 3 seeds. PPO with a learned critic reaches 77–94%, and the RBF neural-network adaptive controller of [2] reaches 62%. GRPO settles fastest and is most robust to gusts, while PPO needs 100× fewer samples and generalizes better to unseen plants.

## I. Problem and Environment

Dynamics. The state is $q = [x, \theta_1, \theta_2]$: trolley position, hook angle and payload angle. The Euler–Lagrange equations $M(q)\ddot{q} = F(q, \dot{q}) + \tau$ are taken from [2], with viscous damping. They are integrated with RK4 at $\Delta t = 0.02\,\mathrm{s}$ and are valid for large angles. A horizontal gust $F_d$ on the payload enters through $\tau = J^\top F_d = F_d[1,\, L_1 \cos\theta_1,\, L_2 \cos\theta_2]^\top$. Tests check energy conservation without damping (drift $< 10^{-3}$) and that the energy change equals the gust's work.

Task. An episode lasts 12 s (600 steps). Its scenario is sampled at reset:

- payload $m_2 \in [0.5, 4]\,\mathrm{kg}$ and cable lengths $L_1 \in [0.6, 1.4]\,\mathrm{m}$, $L_2 \in [0.4, 1.2]\,\mathrm{m}$;
- a target at a distance of $\pm[1, 4]\,\mathrm{m}$;
- initial sway up to $\pm 30°$;
- up to three gusts of 2–6 N lasting 0.2–0.5 s, occurring during transport (the final 4 s are calm).

The gust schedule is pre-sampled, so the scenario is exactly replayable. End-stops at $|x| = 6\,\mathrm{m}$ stop the trolley. During training a curriculum widens all ranges from 30% to 100% over the first 30% of training.

## II. RL Formulation

State space $\mathcal{S} \subset \mathbb{R}^{13}$. The observation, with every entry normalized, is

$$s = [e, \dot{x}, \sin\theta_1, \cos\theta_1, \dot{\theta}_1, \sin\theta_2, \cos\theta_2, \dot{\theta}_2, e_p, u_{t-1}, m_2, L_1, L_2],$$

where:

- $e = x - x^*$ is the trolley's position error;
- $e_p$ is the payload's horizontal position error;
- $u_{t-1}$ is the previous action;
- $(m_2, L_1, L_2)$ are the plant parameters, as measured by a load cell and a rope encoder. An ablation hides them (10-D).

Sin/cos encoding avoids angle wrap-around.

Action space $\mathcal{A} = [-1, 1]$ (continuous). The trolley force is $F = 30u$ N.

Reward. With settled indicator $\sigma_t$ ($|e| < 5\,\mathrm{cm}$, $|\theta_{1,2}| < 2°$, $|\dot{x}| < 5\,\mathrm{cm/s}$):

$$\begin{aligned} r_t = -\big[&|e| + 0.05\dot{x}^2 + 2(\theta_1^2 + \theta_2^2) + 0.1(\dot{\theta}_1^2 + \dot{\theta}_2^2) \\ &+ 0.01u^2 + 0.1(u_t - u_{t-1})^2\big] \\ &- 0.1(1 - \sigma_t) + \sigma_t - 10\,\mathbb{1}_{\text{end-stop}}. \end{aligned}$$

The linear $|e|$ term gives a constant pull toward the target, and together with the per-step time cost it rewards fast transport. The sway and rate terms penalize the coupled oscillations. The $\Delta u$ term asks for smooth force. The settle bonus rewards finishing with the payload at rest.

## III. Algorithms

GRPO for control (ours). Each iteration samples $N = 64$ scenarios ("prompts"). For each one it rolls out $G = 32$ trajectories ("outputs") of a Gaussian MLP policy (2×128, tanh), giving 2048 cranes simulated in parallel in PyTorch on the GPU. There is no critic. Advantages are group-relative, and we compare three variants:

- outcome [1]: $\hat{A}_i = (R_i - \bar{R})/\sigma_R$;
- process [1]: the suffix sum of group-normalized rewards;
- rtg (ours, default): the discounted return-to-go, normalized across the group at each timestep, $\hat{A}_{i,t} = (G_{i,t} - \overline{G_{\cdot,t}})/\sigma_{G_{\cdot,t}}$. This is a per-time baseline suited to dense rewards.

Because all members of a group share the scenario, the baseline removes the variance caused by the plant and the disturbances, which is the job a learned critic would do in PPO. The loss is the GRPO objective (clipped ratio, $\epsilon = 0.2$) with the unbiased KL estimator $\frac{\pi_{ref}}{\pi} - \log\frac{\pi_{ref}}{\pi} - 1$ ($\beta = 0.04$). The reference policy is refreshed every 10 iterations (iterative GRPO).

RL baseline: PPO [3]. PPO uses the same environment, curriculum and validation set as GRPO. It has a learned critic with GAE ($\lambda$=0.95), $\gamma$=0.995, reward normalization, 12 CPU environments, the same 2×128 tanh network, and 3M steps.

TABLE I: GRPO vs PPO vs RBF adaptive control (100 episodes per condition; best in bold). $t_s$ is averaged over settled episodes. OOD: $m_2$=5 kg, $L_1$+$L_2$=2.8 m.

| | Success (%) | | | $t_s$ (s) | | |
|---|---|---|---|---|---|---|
| Condition | GRPO | PPO | RBF | GRPO | PPO | RBF |
| Nominal | 100 | 96 | 71 | 7.1 | 8.0 | 9.2 |
| Random | 94 | 94 | 62 | 7.1 | 7.9 | 8.8 |
| Heavy | 100 | 97 | 72 | 6.7 | 7.7 | 8.5 |
| Light | 48 | 56 | 23 | 9.0 | 8.9 | 9.3 |
| Long cable | 73 | 78 | 31 | 8.9 | 8.9 | 10.5 |
| Short cable | 100 | 100 | 75 | 6.4 | 7.4 | 8.6 |
| Strong gusts | 92 | 81 | 48 | 7.0 | 8.3 | 9.0 |
| OOD | 81 | 96 | 56 | 8.2 | 8.2 | 9.9 |

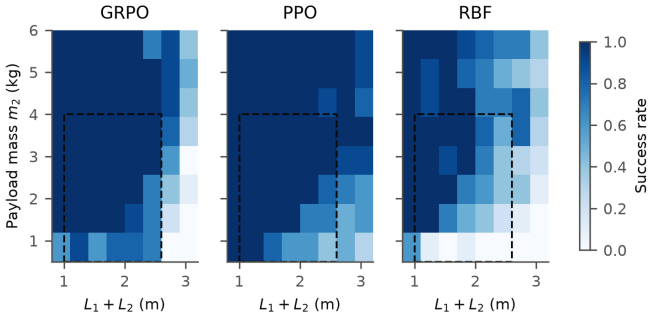



Fig. 1: Success rate over payload mass and cable length (10 episodes per cell). The dashed box is the training range.

Classical baseline: RBF adaptive control [2]. This controller is not learned. Its control law is $\tau = M_0(q)(\ddot{q}_d - K_v\dot{e} - K_p e) + C_0\dot{q} + G_0 - \hat{f}$, built on the nominal model ($m_2$=2, $L_1$=1, $L_2$=0.8) and following a quintic reference.

- RBF compensator: $\hat{f} = W^\top h([e, \dot{e}])$ with 11 Gaussian units. The weights adapt online as $\dot{W} = \gamma\, h\, [e, \dot{e}]^\top PB$, where $PA + A^\top P = -I$.
- Underactuation: only the trolley row of $\tau$ is actuated.
- Gains: with the paper's positive sway gains the sway was not damped (12% success). A grid search gave $K_p$=diag$(5, -22, -22)$, $K_v$=diag$(6, 0.5, 0.5)$, $\gamma$=5.

## IV. Results

Protocol. Each condition uses 100 seeded test scenarios that are never used in training. An episode succeeds if the crane is settled during the final second; settling time $t_s$ is the first instant after which it stays settled. All controllers run on the same scenarios and gust schedules. The table uses seed 0 of each RL method.

GRPO vs PPO. On random scenarios GRPO's three seeds reach 94/93/97% success and PPO's reach 94/84/77%. The group baseline gives markedly lower seed-to-seed variance than a learned critic.

- Where GRPO is better. It settles fastest in every condition (7.1 s vs 7.9 s on random scenarios) and is clearly more robust to doubled gusts (92% vs 81%).
- Where PPO is better. PPO needs 100× fewer samples (3M vs ∼300M steps; Fig. 2). Batched GPU simulation offsets this: GRPO passes 85% validation success in about 3 min on an RTX 3050 Ti laptop GPU. It is smoother (0.5° residual sway, 0.64× GRPO's control energy) and extrapolates better beyond the training range (OOD 96% vs 81%; Fig. 1).
- Why. GRPO's Monte-Carlo, scenario-matched baseline is exact for the scenarios it sees. PPO's value function smooths across plants, which helps it extrapolate.

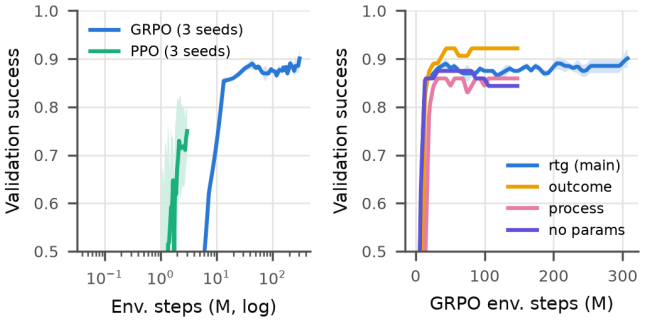



Fig. 2: Validation success on 64 held-out scenarios. Left: GRPO vs PPO, mean ± std over 3 seeds (log steps). Right: GRPO advantage variants and the no-parameter ablation.

RL vs RBF. Both RL agents beat the RBF controller in every condition: 94% vs 62% on random scenarios, and 92/81% vs 48% under doubled gusts. With only the trolley actuated, the computed-traction law reaches the sway only through the inertia coupling of $M_0(q)$, along a fixed reference. Online RBF adaptation barely helps (57% with adaptation vs 63% with $\gamma$=0), because compensating model error cannot create the missing actuation. The RBF controller degrades quickly with long cables and light payloads (Fig. 1).

Ablations (Fig. 2; 100 random test scenarios; ablation runs use 120 iterations, the main run 250).

- Success saturates for every variant (rtg 94%, outcome 96%, process 95%), but the return separates them: rtg 61, outcome 45, process −7.
- The paper's process advantage is worst (32° peak sway). Its suffix sums over 600 dense steps carry a time-dependent offset shared by every trajectory.
- Hiding $(m_2, L_1, L_2)$ keeps 94% success (return 48). No load cell or rope encoder is needed.

## V. Conclusion

Critic-free GRPO transfers from LLM reasoning to physical control. Replaying each scenario across a group gives a low-variance baseline, which yields the fastest-settling and most gust-robust controller here, consistent across seeds. PPO is more sample-efficient and extrapolates better. Both RL methods clearly beat RBF adaptive control.

## References


[1] Z. Shao et al., "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," arXiv:2402.03300, 2024.

[2] H. Qiang, Y. Sun, J. Lyu, D. Dong, "Anti-sway and positioning adaptive control of a double-pendulum effect crane system with neural network compensation," Front. Robot. AI, vol. 8, 639734, 2021.

[3] J. Schulman et al., "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.